MACHINE-CHECKED OBSTRUCTIONS FOR POTENTIAL-BASED APPROACHES TO THE COLLATZ CONJECTURE, WITH CLOSED-FORM BOUNDED BARRIERS

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Abstract. We formalize, in Lean 4 over a single verified graph model — the admissible inverse cylinder tree of the accelerated Syracuse map — an obstruction map for four natural families of "local potential" proof strategies aimed at the divergence half of the Collatz problem, together with the strongest closed-form positive results those obstructions permit. The negative results are theorems, not heuristics: the per-block induction route is closed (its naive count form is provably equivalent to the barrier it was meant to prove, and the crux of its non-circular form is refuted by an explicit astronomical one-edge ladder); no congruence invariant separates start from goal at any finite modulus (every unit cylinder is reachable); no potential with a Q-independent 3-adic continuity modulus certifies the barrier uniformly in Q, in any codomain; and the linear bit-length class, which certifies the barrier in closed form exactly up to a bit criterion, dies at the very next step — and enriching it with singular poles at the two attractors does not move the wall by a single step (an LP-duality argument whose rational Farkas certificates are themselves checked in Lean). The positive results include the closed-form bit-length potential, and a kernel-checked formalization of Eliahou's continued-fraction cycle reduction. The residual difficulty is isolated as a single named phenomenon — the archimedean Sturmian carry process of rotation number $\log_2 3$ — for which the function-field analogue over $\mathbb{F}_2[x]$, here formalized end-to-end, serves as a control experiment. Every declaration cited depends only on the three standard Lean axioms (machine-audited). The Collatz conjecture itself remains, of course, wide open; the contribution is a machine-checked map of where its difficulty does and does not live.

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The Lean formalization was developed in an extended human–AI collaboration with the assistant Claude (Anthropic); see §10 for the methodology and the division of labour, and the Acknowledgements. The artifact: `https://github.com/benetandujar72/collatz-admissible-tree`.

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1. Introduction

The Collatz (or $3n + 1$) conjecture asserts that iterating $n \mapsto n/2$ (n even), $n \mapsto 3n + 1$ (n odd) from any positive integer eventually reaches 1. It famously splits into two independent halves: *no nontrivial cycles*, and *no divergent orbits*. Both are open; for the state of the art see Lagarias's surveys [12, 13], Tao's almost-everywhere result [23], the cycle exclusions of Simons–de Weger [19] and Hercher [8], and the computational verification frontier [1, 17].

This paper reports on a formalization programme (sessions S189–S244 of an ongoing human–AI collaboration) that attacks the problem from the *inverse* direction: instead of following orbits forward, it studies the tree of 3-adic cylinders that can reach a fixed target cylinder, and asks for *local potential functions* certifying that "cheap" inverse paths do not exist. The programme produced positive results — closed-form barriers — but its principal output is sharper and, we believe, more useful: a *machine-checked obstruction map* showing that four natural families of potential-shaped strategies on this graph fail, for identified, theorem-level reasons.

Why publish negative results? Because rigorous obstruction knowledge redirects effort: a verified "this cannot work, and here is the certificate" spares a community years of unrecorded failed attempts, as recent machine-assisted refutations in extremal combinatorics have illustrated. The Collatz literature is dense with potential and invariant attempts; the map below makes a large class of them obsolete *as theorems*, each failure carrying a finite, independently re-checkable witness.

Contributions. All results are formalized in Lean 4 [14] over Mathlib [15]; the artifact builds with *zero* `sorry`. Headline theorems (Lean names in typewriter font; the axiom audit is Appendix A):

(1) **The induction route is closed** (§3): the naive per-block count conjecture is provably equivalent to the barrier it was meant to imply (`S202_one_edge_count_conjecture_iff_defect_barrier`); the corrected, non-circular reduction has a single crux (`BlockBoundaryExists`), and that crux — together with its first-m-precise variant — is *refuted* for all $m, Q \geq 1$ by an explicit astronomical one-edge ladder built in closed form (`ladder_exists`; Theorem 3.4).

(2) **No congruence invariant exists** (§4): every unit residue class modulo every power of 3 is reachable from the start (`invReachable_units`); separating invariants at any finite modulus are therefore impossible.

(3) **No Q-uniformly continuous potential exists, in any codomain** (§5): the two *attractor identities* $4(y-1) = 3(c-1)$, $2(y+1) = 3(c+1)$ force any 3-adic continuity modulus ω of a certifying potential to obey $m \leq \omega(22 + Q)$ (`no_decaying_modulus_corrector`), so no modulus independent of Q — in particular no local constancy — can serve.

(4) **A closed-form positive frontier, with an exact machine-checked wall** (§6): the potential $\Phi_{\text{bit}} = \lfloor(\text{size}(c) - 1)/4\rfloor + j - Q$ proves the bounded κ-barrier under the bit criterion $4(m+Q) + 1 \leq \text{size(start)}$ (`kappa_bounded_barrier_bitlen`, faithful instances exactly tight at 73 and 105 bits); and *no* member of the linear class survives one step further (`no_linear_size_potential_m1_Q8`), while the linear relaxation of Φ_{bit} is a member of the class satisfying the contract at the step before (`linear_size_potential_m1_Q7`): the wall is certified on both sides. Enriching the class with singular poles at the two attractors does not move the wall (`no_hybrid_potential_m1_Q8`); the Farkas certificates are rational, at most six rows long, and checked in Lean.

(5) **The cycle pipeline, formalized** (§7): unconditional non-existence of Syracuse cycles of period ≤ 4; the bilateral window $m \log_2 3 < A < 2m$; a one-inequality *master corridor* valid without any cycle hypothesis; and Eliahou's continued-fraction reduction [6] — the elementary core of the Simons–de Weger–Hercher programme — with kernel-checked instances (Syracuse periods < 359 from verification up to 2^{20}; periods $< 16\,266$ from 2^{29}), connected to the literature's verification statements by an explicit bridge.

(6) **The function-field control experiment, formalized end-to-end** (§8): the Collatz analogue on $\mathbb{F}_2[x]$ of Hicks–Mullen–Yucas–Zavislak [9] is proven in Lean (`collatz_F2X`) — to our knowledge

its first formalization — with the exit from degree plateaus governed by the identity $x(T^2f+1) = (x+1)(f+1)$, the trailing-coefficient argument of Snapp–Tracy [21]. The same four local laws that are exact over $\mathbb{F}_2[x]$ are lossy over $\mathbb{Z}$; the difficulty of the integer problem is thereby exhibited theorem-against-theorem inside one artifact.

What this paper does not claim. Nothing here proves or claims progress on the Collatz conjecture itself. The barriers of §6 are finite statements about a specific cylinder tower; the obstructions say that a family of proof *shapes* cannot work on this model. We consider the honest delimitation of scope a feature of the machine-checked format: every theorem below means exactly what its Lean statement says, no more.

2. The verified model

2.1. **The accelerated map and admissible classes.** For odd $n \in \mathbb{N}_{>0}$ let $\mathrm{Syr}(n) = (3n+1)/2^{\nu_2(3n+1)}$ be the accelerated Syracuse map. The inverse of an odd step is $n \mapsto (2^t n - 1)/3$, which is an integer coprime to 3 iff $2^t n \equiv 4$ or 7 (mod 9). For an *admissible* class $\alpha \in X_9 = \{1, 2, 4, 5, 7, 8\}$ (the units modulo 9) let $\tau(\alpha)$ be the least $t \geq 1$ with $2^t\alpha \equiv 4$ or $7 \pmod 9$; this gives the table

$$\tau: \quad 1 \mapsto 2, \quad 2 \mapsto 1, \quad 4 \mapsto 2, \quad 5 \mapsto 3, \quad 7 \mapsto 4, \quad 8 \mapsto 1$$

(`tau`, `InX` in `AdmissibleBasic.lean`). Note that τ is attached to the class of the value being pre-imaged (the successor); it is the doubling exponent of the inverse branch, not $\nu_2(3\alpha+1)$. Throughout, τ is this table value, with $\tau \leq 4$ on X_9 (`tau_le_four`; classes $0, 3, 6$ never occur on odd orbits).

2.2. **The inverse cylinder graph.** Fix a base precision R. Vertices are pairs $v = (j, c)$ with $j \in \mathbb{N}$ (the number of *zero-steps* consumed) and c a residue class modulo 3^{R+j} (`InvVertex`). There are two edge families (`InvEdgeOne`, `InvEdgeZero`):

$$\text{one-edge: } (j,c) \to (j,c') \quad \text{iff} \quad c' \in X_9,\ c \equiv 2^{\tau(c')}c' \pmod{3^{R+j}},$$

$$\text{zero-edge: } (j,c) \to (j{+}1,c') \quad \text{iff} \quad c' \in X_9,\ 3c+1 \equiv 2^{\tau(c')}c' \pmod{3^{R+j+1}}.$$

One-edges invert a pure doubling block; zero-edges invert one odd step and *refine the modulus by one* 3*-adic digit.* The *κ-weights* (`InvKappaPreciseEdge`) are

$$\kappa(\text{one-edge}) = +1, \qquad \kappa(\text{zero-edge}) = \begin{cases} -1 & \tau(c') = 1, \\ 0 & \tau(c') \geq 2, \end{cases}$$

so that along any path $\sum \kappa = \#\text{one-edges} - \#\{\tau{=}1 \text{ zero-edges}\}$, the combinatorial shadow of the *defect* $D(u) = A(u) - 2q(u)$ of the corresponding forward word (`InvPathCounts.weight_lower_bound`).

2.3. **Words, the translation identity, and the S202 direction.** An *admissible word* u over $\{\texttt{0}, \texttt{1}\}$ encodes a composition of inverse Collatz steps: `1` multiplies by the forced power 2^τ, `0` applies $n \mapsto (2^\tau n - 1)/3$. The accumulated action is affine: a word with parameters (A, q, B) — $A = \sum \tau$, q the number of zeros — maps $n_0 \mapsto (2^A n_0 - B)/3^q$, and words run from $n_0 = 1$ satisfy the *translation identity* $3^q n + B = 2^A$ (`evalWord`, `S202_translation_identity`).

The programme's organizing example is the 26-letter word W_{202} with $(A, q) = (43, 22)$, $B_{202} = 919\,447\,060\,349$ and final value $n = 251$. Its *defect* is $D = A - 2q = -1 < 0$: it refutes the naive monotonicity hope $A \geq 2q$ and opens the programme's asymptotic slope question ($\liminf A/q \geq 2$? — subcritical slope along admissible words would be the inverse-tree shadow of slow forward growth). The fixed point of its affine action,

$$x^* \;=\; \frac{B_{202}}{2^{43} - 3^{22}} \;\approx\; 0.105 \;\notin\; \mathbb{Z},$$

is a *virtual cycle*: no integer realizes it (there is no actual 43-double, 22-odd-step cycle), but 3-adically it is a unit, and its 3-adic expansion defines a coherent tower of cylinders

$$C_m \;=\; \{\, n \equiv a^* \pmod{3^R} \,\}, \qquad R = 22m + 2,$$
$$a^* \;\equiv\; B_{202}\left(2^{43} - 3^{22}\right)^{-1} \pmod{3^R},$$

heuristically the integers on which W_{202} can be applied backward m times within the cylinder's precision (each application consumes the word's $q = 22$ digits of 3-adic precision; the $+2$ is the mod-9 base). Pleasingly, at $m = 1$ the canonical residue is exactly $a^* = 1 + 3^{22}$ (`aS202_decomp`), of bit-size 35; this agreement holds exactly modulo 3^{26} and fails modulo 3^{27}.

The tower C_m is thus the canonical *cycle-resonant direction* of the inverse tree: the region where cheap (low-defect) inverse returns to 1 would concentrate if they existed. The corpus records the resulting dichotomy as the (open) *S202 alternative* (`S202Cylinders.lean`): either every C_m is accessible from 1 at defect cost $o(m)$ — giving admissible families of subcritical slope $\liminf A/q \leq 43/22 < 2$ — or accessibility fails infinitely often and a new 3-adic obstruction governs the admissible inverse cover. The barriers studied below are the quantitative, budget-bounded face of the second horn: they assert that within a zero-step budget Q there are *no* cheap returns along this tower. We emphasize the honest scope: C_m is a model family — the hardest natural direction first — and barriers on it are neither necessary nor sufficient for Collatz; their role is that every potential or induction strategy on the inverse tree must already succeed here, which is what makes the obstruction map of §§3–6 informative for strategy design at large.

2.4. **The start, the goal, and the barrier.** In the artifact the graph-level start is $\mathrm{Start}(m) = (0,\; (1 + 3^{22}) \bmod 3^R)$ (`InvStart`, `aS202`). For $m = 1$ this is the tower residue $a^* \bmod 3^R$; for $m \geq 2$ it is the $m = 1$ residue viewed at precision $22m+2$ — the corpus's "Caveat C". The faithful residues a_m^* (`aS202_at`, `InvStartAt`) are used for the $m = 2, 3$ instances of §6; the obstruction theorems of §§3–5 are stated for $\mathrm{Start}(m)$, and their proofs use only $a \equiv 1 \pmod 9$ and $\nu_3(a-1) = 22$, properties shared by every a_m^*, but they are not formalized for the faithful tower. A vertex is a *goal* if its cylinder contains 1, i.e. $c \equiv 1 \pmod{3^{R+j}}$ (`InvVertex.IsGoal`).

Definition 2.1 (the bounded κ-barrier; `S202_kappa_precise_barrier_bounded`)**.** For $m, Q \in \mathbb{N}$, every κ-weighted inverse path from $\mathrm{Start}(m)$ to a goal with at most Q zero-edges has total $\kappa \geq m$.

The semantic content on the forward side is the *slope barrier*: every admissible word u with $q(u) \leq Q$ landing in the cylinder of $\mathrm{Start}(m)$ has defect $D(u) \geq m$ (`S202_slope_barrier_from_kappa_precise_potential_bounded`, via the verified forward–inverse correspondence `S212_forward`). Informally: there are no cheap inverse returns along this resonant 3-adic direction. These are finite, decidable-in-principle statements; their interest is as the minimal non-trivial quantitative expansion property the inverse-tree programme needs, and as the target all potential strategies below aim at.

Theorem 2.2 (the potential method; `S202_kappa_precise_barrier_bounded_from_potential`)**.** *Let Φ be any function from vertices to $\mathbb{Z}$ (or $\mathbb{Q}$, $\mathbb{R}$) such that* (E) $\Phi(v) \leq \kappa(e) + \Phi(v')$ *for every edge $e : v \to v'$ in the slice $j \leq Q$;* (G) $\Phi(g) \leq 0$ *for every goal g with $g.j \leq Q$; and* (S) $\Phi(\mathrm{Start}(m)) \geq m$. *Then the bounded κ-barrier holds at (m, Q).*

Everything in §§3–6 is about which Φ can exist.

2.5. **The mod-9 rigidity.** Two structural facts used throughout: the source class of every one-edge lies in $\{4, 7\} \bmod 9$ (`oneEdge_source_mod9_in_four_seven`; equivalently $2^{\tau(\alpha)}\alpha \equiv \{4, 7\} \pmod 9$ for all $\alpha \in X_9$, `two_pow_tau_mul_mod9`) — so every one-edge source has $\nu_3(c-1) = 1$; and the *deep-jump trap*: a one-edge lands on a target with $\nu_3(c'-1) \geq 2$ — equivalently its source satisfies $c \equiv 2^\tau \pmod 9$ — *iff* the target is $\equiv 1 \pmod 9$, in which case the source class is 4 and $\tau = 2$ (`oneEdge_deepjump_iff`, `oneEdge_deepjump_source_four`). The trap will reappear, unplanned, inside the LP certificates of §6 — the two obstruction families are one mechanism.

3. Obstruction I: the induction route is closed

The natural uniform-in-m strategy is induction on the precision level: cut each level-$(m+1)$ path at a distinguished vertex whose prefix projects to an m-level path (apply the induction hypothesis: $\kappa_1 \geq m$) and whose suffix contributes $\kappa_2 \geq 1$. The formalization makes the folklore difficulty exact, in three steps.

Theorem 3.1 (the naive count form is circular; `S202_one_edge_count_conjecture_iff_defect_barrier`)**.** *The programme's first formulation of* *the per-block increment — the existence, for every word landing in the cylinder, of counts $n_{\text{one}}, n_{\text{neg}}$ with $n_{\text{one}} - n_{\text{neg}} \le D$ and $m \le n_{\text{one}} - n_{\text{neg}}$ — is logically equivalent to the slope barrier $D \ge m$ it was meant to imply.*

The corrected reduction pins the counts to the actual edges of the path (`InvPathCounts`); its split hypothesis `KappaPathSplit` carries strictly more information than the $(m{+}1)$-barrier (`kappaPathSplit_yields_piecewise_bounds`). (For a split point subject to no requirement beyond $\kappa_1 \ge m$, $\kappa_2 \ge 1$, a two-line discrete intermediate-value argument on the $\{-1, 0, +1\}$-valued partial sums gives the converse as well, so that split is again equivalent to the barrier; this elementary remark is not formalized.) A useful reduction must therefore couple the split point to the m-level structure:

Theorem 3.2 (the reduction, and its crux; `kappaPathSplit_of_block`, `uniform_kappa_barrier_of_block`, `uniform_slope_barrier_of_block`)**.** *Let `BlockBoundaryExists`(m, Q) be the statement: every level-$(m{+}1)$ κ-path from the start to a goal of depth $\le Q$ admits a split point* mid *that is m-precise (its prefix projects onto an m-goal) and whose suffix has $\kappa_2 \ge 1$. Then a base barrier at level m_0 together with `BlockBoundaryExists` at every level $\ge m_0$ yields the bounded κ-barrier — and hence the slope barrier $D \ge m$ — at every level $m \ge m_0$.*

Theorem 3.2 is only useful if its crux can be supplied. It cannot:

Theorem 3.3 (the unconditioned increment is false; `outerBlockIncrement_false`)**.** *The statement "every suffix from a projected m-goal contributes $\kappa_2 \ge 1$" is refutable outright (take the root vertex and the empty suffix).*

Theorem 3.4 (the crux and its canonical strengthening are false)**.** *For all $m, Q \ge 1$, `BlockBoundaryExists`(m, Q) is false (`not_blockBoundaryExists`); a fortiori the first-m-precise-vertex device `FirstMPrecisionSuffixPositive`(m, Q) is false (`not_firstMPrecisionSuffixPositive`). Theorem 3.2 therefore reduces the barrier to a false hypothesis: the induction route, in every formulation the programme found, is closed.*

The refutation is constructive and, we think, the most instructive object of the negative half: a path consisting of one $\tau \ge 2$ zero-edge (weight 0) followed by an astronomical *one-edge ladder* alternating between the classes 4 and 7 mod 9 (`ladder_exists`; the "mod-6 shield" `mod6_of_two_pow_mod9_seven` keeps every rung non-m-precise), ending with a single deep one-edge into the goal $(1, 1)$ (`deep_edge`). The ladder's length is governed by $\operatorname{ord}(2 \bmod 3^{R+1}) = 2 \cdot 3^R$ (`orderOf_two_zmod_three_pow`) — about 10^{22} rungs already at the $m = 2$ precision — yet it exists *in closed form*: no enumeration appears anywhere in the proof. On this path every m-precise split point forces a suffix with $\kappa_2 = 0 < 1$, refuting the crux. Notably, the same order-of-2 computation also disposes of all closed one-edge walks (`closed_oneWalk_weight_ge_order`): the graph has no short κ-cheap cycles to exploit.

Remark 3.5. The barrier itself survives Theorem 3.4: on the ladder path the prefix already pays $\kappa_1 \geq m$, and the refuted object is the split *predicate*, not the inequality. What dies is every decomposition whose split point is required to be m-precise; a viable induction would have to locate the per-block increment elsewhere, or the route must be abandoned.

4. Obstruction II: no finite-modulus invariant

A second classical hope is a congruence invariant: a union of residue classes modulo 3^t (or any modulus) containing the start, closed under edges, and avoiding the goal.

Theorem 4.1 (reach = units; `invReachable_units`)**.** *For every $m \geq 1$, every $t \geq 1$ and every z coprime to 3, some vertex reachable from* $\mathrm{Start}(m)$ *satisfies $v.c \equiv z \pmod{3^t}$.*

The proof is an induction on precision: each zero-edge contributes exactly one new 3-adic digit, and the lifting lemma `invReachable_zero_lift` (driven by `two_pow_tau_mul_mod9`) realizes any prescribed next digit. Consequently no edge-closed union of residue classes modulo 3^t — for any t — separates start from goal. Even the "trap" cylinder of the deep-jump analysis is reached at every precision (`invReachable_hits_trap`).

5. Obstruction III: no continuous potential, in any codomain

Could Φ be 3-adically *continuous* — depending, with decaying influence, on deeper and deeper digits of c? For $\mathbb{Z}$-valued Φ, continuity means local constancy: a modulus ω vanishing beyond some depth t_0, which Theorem 5.2 below excludes as soon as $22 + Q \geq t_0$. The general case rests on two exact identities.

Proposition 5.1 (the attractor identities; `tau2_attractor`, `tau1_attractor`, `tau2_attractor_nu3`, `tau1_attractor_nu3`)**.** *Along a $\tau = 2$ zero-edge with target y, $4(y-1) = 3(c-1)$, so $\nu_3(y-1) = \nu_3(c-1)+1$; along a $\tau = 1$ zero-edge, $2(y+1) = 3(c+1)$, so $\nu_3(y+1) = \nu_3(c+1)+1$.*

Thus the τ=2, class-1 chain converges 3-adically to the goal 1 while every edge on it has $\kappa = 0$, and the τ=1 filament converges to -1 paying $\kappa = -1$ per step. Moreover the start sits *on* the $+1$ attractor shell: $\nu_3(a^* - 1) = 22$ (`nu3_aS202_sub_one`; indeed $a^* - 1 = 3^{22}$ exactly at $m = 1$). Following the chain from the start for Q steps (`attractor_chain`) yields:

Theorem 5.2 (no decaying-modulus corrector; `no_decaying_modulus_corrector`)**.** *Let $m \geq 1$ and let Φ : vertices $\to \mathbb{R}$ satisfy the edge inequality (E) on* all *edges of the level-m graph (not only the slice), (G) on the slice $j \leq Q$, and (S), together with a 3-adic continuity modulus ω: equal j and $c \equiv c' \pmod{3^t}$ imply $\Phi(v) - \Phi(v') \leq \omega(t)$. Then $m \leq \omega(22+Q)$.*

Read at fixed m: a Φ with continuity modulus ω can certify (m, Q) only if $\omega(22 + Q) \geq m$ — its modulus must still be at least m at depth $22 + Q$, exactly the depth reached by Q zero-edges from the start's 3^{-22}-shell. Hence no single decaying modulus serves a fixed m for all Q, and no Q-independent continuity assumption is compatible with the contract, in any codomain. (At a fixed (m, Q) a Φ with $\omega(22 + Q) \geq m$ is not excluded: whenever the barrier is true its exact value function is such a Φ.) Numerical companions (Appendix C) quantify the failure: on certified finite regions, the empirical Hölder constant required at exponent α scales as $3^{\alpha(R+Q+\text{const})}$ — astronomically slack — and a Green-type singular part centred on the natural pole set does not repair it (the "interior" depth profile is robust to enlarging the pole family to $2^s\{\pm1, \pm4, \pm a^*\}$, $s \leq 8$). The tropical (min-plus) variant of the same idea is refuted by an explicit six-edge quotient-graph witness, and the optimal-path language is the period-one $\tau{=}1$ filament rather than a Sturmian word — the rotation number $\log_2 3$ enters globally, not locally (tool-level verdicts, tier T2/T3 of §10).

6. The positive frontier, and its exact wall

6.1. **The closed-form barrier.** The two integer *size laws* — with $\mathrm{size}(n)$ the bit length —

$$\mathrm{size}(2^\tau n) = \mathrm{size}(n) + \tau, \qquad \mathrm{size}(3n + 1) \geq \mathrm{size}(n) + 1$$

(`size_two_pow_mul`, `size_three_mul_add_one`; the second is an inequality, and *all later difficulty lives in its slack*) give edge estimates for the canonical residues (`oneEdge_size_le`, `zeroEdge_size_le`) and hence:

Theorem 6.1 (the bit-length potential; `PhiBitlen`, `kappa_bounded_barrier_bitlen`, `kappa_bounded_barrier_bitlen_from`)**.** *Define* $\Phi_{\mathrm{bit}}(v) = \lfloor(\mathrm{size}(v.c \bmod 3^{R+j}) - 1)/4\rfloor + j - Q$. *Then* Φ_{bit} *satisfies (E) and (G) unconditionally, and (S) holds whenever*

$$4(m + Q) + 1 \ \leq\ \mathrm{size}(\textit{start residue}) + 4\, j_{\textit{start}}.$$

Hence the bounded κ-barrier holds, in closed form and with no search, under the bit criterion.

Instances: at $m = 1$ (35 bits) the criterion gives $Q \leq 7$ (`kappa_barrier_m1_Q7`), subsuming the per-instance BFS certificates; a uniform corollary covers all $m + Q \leq 8$ in one stroke (`kappa_barrier_of_sum_le_eight`); and for the *faithful* per-m starts a^*_m (the Caveat-C-corrected tower, `aS202_at`) the criterion is *exactly* tight: $m = 2$, $Q \leq 16$ with $4(2{+}16){+}1 = 73 = \mathrm{size}(a^*_2)$, and $m = 3$, $Q \leq 23$ with $4(3{+}23){+}1 = 105 = \mathrm{size}(a^*_3)$ (`kappa_barrier_at_m2_Q16`, `kappa_barrier_at_m3_Q23`). All of these are kernel-checked on stored numerals; no compiled evaluation (`native_decide`) remains in this chain.

6.2. **The wall, machine-checked on both sides.** Is the bit criterion an artifact of Φ_{bit}, or the boundary of its whole class? We answered this with a linear program over the feature coefficients, solved by constraint generation (the slices at the wall scales are far beyond practical enumeration, but the LP has five unknowns). The working set consists of rows harvested by walks on the graph together with explicit in-slice edge families: the contract (E) quantifies over *every* edge of the slice, reachable from the start or not, and the decisive rows turn out to be unreachable ones. Every infeasibility verdict carries an exact rational Farkas certificate — finitely many true rows whose nonnegative combination is contradictory — re-verified independently against the raw modular arithmetic; on the feasible side the witness is an explicit member of the class. The $m = 1$ results were then formalized:

Theorem 6.2 (the linear class: alive at $Q = 7$, dead at $Q = 8$; `linear_size_potential_m1_Q7`, `no_linear_size_potential_m1_Q8`). *The linear relaxation of* Φ_{bit}*,* $\Phi(v) = -\frac{29}{4} + j + \frac{1}{4}\,\mathrm{size}(v.c \bmod 3^{24+j})$*, satisfies (E), (G), (S) at* $(m, Q) = (1, 7)$ *(directly from the size laws). No* $\theta_0, \theta_1, \theta_2 \in \mathbb{Q}$ *make* $\theta_0 + \theta_1 j + \theta_2\,\mathrm{size}(v.c \bmod 3^{24+j})$ *satisfy them at* $(1, 8)$*: the certificate has four rows — start, goal, one* $\tau{=}1$ *zero-edge, one* $\tau{=}4$ *one-edge — with multipliers* $(2, 2, 16, 17)/37$*.*

The wall of the linear class thus sits *exactly* at the bit criterion, certified on both sides. (Since the rows of the slice at Q are among those at $Q + 1$, infeasibility is monotone in Q.)

Theorem 6.3 (attractor poles do not move the wall; `no_hybrid_potential_m1_Q8`). *Enrich the class with the two singular features* $\nu_3(c-1)$ *and* $\nu_3(c+1)$ *(poles at the attractors of §5, the zero residue valued at full depth). No* $\theta \in \mathbb{Q}^5$ *satisfies the contract at* $(1, 8)$*: the enriched class dies at the same step as the linear one. The six-row certificate has multipliers* $(1798, 1798, 14877, 13888, 496, 564)/33421$*; its decisive row is the goal's own one-edge predecessor* $(8, 4) \to (8, 1)$ *— the mod-9 trap of §2 sitting immediately next to the goal — along which* $\nu_3(c-1)$ *jumps from* 1 *to the full depth* 32 *at cost* $\kappa = +1$*, capping any pole weight.*

(An earlier six-row certificate at $(1, 9)$, `no_hybrid_potential_m1_Q9`, remains in the artifact; it is implied by Theorem 6.3 by monotonicity.)

The numerical sweep extends the picture (tier T2: exact certificates at every dead point; "alive" means the LP over the harvested rows is feasible and, at the stated Q, the linear relaxation of Φ_{bit} is a witness):

class	$m = 1$ (35-bit start)	$m = 2$ (73-bit faithful start)
linear $(1, j, \mathrm{size})$	alive $Q \le 7$ / dead $Q \ge 8$	alive $Q \le 16$ / dead $Q \ge 17$
+ poles $\nu_3(c \mp 1)$	alive $Q \le 7$ / dead $Q \ge 8$	alive $Q \le 16$ / dead $Q \ge 17$

The mechanism is uniform: any weight on $\nu_3(c-1)$ must be paid along the goal-adjacent trap edge $(j, 4) \to (j, 1)$, and any weight on $\nu_3(c+1)$ along the τ=2 zero-edge leaving the deep -1 shell, so the poles add nothing to the linear budget. (A first version of the separation oracle harvested only start-reachable edges and reported the enriched class alive one step longer; the explicit in-slice families corrected this — a reminder that a survivor verdict is never a feasibility proof.) Honest scope: these theorems kill *feature-linear* classes against the potential contract of Theorem 2.2; a non-linear integer potential always exists when the barrier is true (the exact value function — e.g. the certified $(1, 8)$ barrier itself, `Cert_m1_Q8_S216_Barrier`). The content is that no *closed form* of these shapes reaches it.

7. The cycle half: Eliahou's reduction, formalized

The cycle half of the programme formalizes Eliahou's classical continued-fraction reduction [6] — the elementary core underlying the later cycle exclusions of Simons–de Weger [19] and Hercher [8], which add Steiner's local-minima decomposition and Baker-type bounds — in elementary, kernel-checkable form. Throughout, a *Syracuse cycle of period* m is $\mathrm{Syr}^{[m]}(a_0) = a_0$ on odd $a_0 > 1$, $m \geq 1$; m counts the odd elements of the cycle and is *not* the m of the "m-cycle" terminology of [19, 8], which counts local minima.

Theorem 7.1 (small periods, unconditionally; `syr_no_nontrivial_fixedpoint` through `syr_no_nontrivial_4cycle`). *There is no nontrivial Syracuse cycle of period* $m \leq 4$.

Theorem 7.2 (the bilateral window; `syr_cycle_window`). *Every nontrivial Syracuse cycle of period* m *satisfies* $m \log_2 3 < A < 2m$*, where* $A = \sum_{i<m} \nu_2(3a_i + 1)$.

The upper half comes from the multiplicative telescope $2^A \prod a_i = \prod(3a_i + 1)$ (`syr_cycle_mul_equation`) and $3a + 1 \leq 4a$; it needs no real analysis. A by-product worth recording is a *master corridor* requiring no cycle hypothesis at all (`syr_master_corridor`):

$$3^q\, a_0 \;<\; 2^{A_q}\, \mathrm{Syr}^{[q]}(a_0) \;\leq\; 4^q\, a_0 \qquad (a_0 \geq 1,\ q \geq 1),$$

one inequality whose two shadows are the cycle window and the trivial growth baseline.

The reduction machinery then follows the literature's shape: the gap identity for $|2^A - 3^m|$ (`syr_cycle_gap_eq`), the coupling of cycle elements to a *verification frontier* (`SyrVerifiedUpTo` as an explicit hypothesis; `syr_cycle_elements_above_frontier`, `syr_cycle_gap_mul_frontier_lt`), and the continued-fraction *mediant exclusion*: for a unimodular convergent pair $L/\Lambda < \log_2 3 < U/\Upsilon$, the pure-$\mathbb{N}$ identity $m = \Lambda(Um - A\Upsilon) + \Upsilon(A\Lambda - Lm)$ (`mediant_period_bound`) turns the window into a period bound (`syr_mediant_exclusion`).

Theorem 7.3 (kernel-checked instances; `syr_no_cycle_below_359`, `syr_no_cycle_below_16266`, `collatz_no_cycle_below_359`, `collatz_no_cycle_below_16266`). *Conditional on orbit verification up to* 2^{20}

(resp. 2^{29}), there is no nontrivial Syracuse cycle of period below 359 *(resp.* 16 266*), via the convergent pairs* $84/53 < \log_2 3 < 485/306$ *and* $1054/665 < \log_2 3 < 24727/15601$*. The same conclusions hold from verification of the standard (full) Collatz map, through the bridge* `syrVerified_of_collatzVerified`*.*

The 16 266 instance involves ∼150 000-digit integers; a fuel-based binary-powering function (`fastPowAux`, `fastPow_eq`) makes the comparisons kernel-reducible at logarithmic recursion depth, so the whole pipeline is checked by the proof kernel with no compiled oracle. Convergent pairs are now mechanical; the ceiling of the method — closing *all* periods — is the open archimedean question (§8), for which the artifact provides a named socket (`EffectiveLinearForm`): any future effective linear-forms-in-logarithms bound plugs in as a hypothesis, not an axiom. (For comparison, Eliahou's method run against today's verification frontier excludes every nontrivial cycle with fewer than $7.2 \cdot 10^{10}$ odd elements [8, §1]; Hercher's headline "no m-cycles with $m \leq 91$" bounds the number of local minima and rests on linear forms in logarithms. Our instances demonstrate the verified pipeline at deliberately small frontiers.)

8. The difficulty, named — and the function-field control

8.1. **One object behind four walls.** The residual obstructions of §§3–7 are not four problems but one. The slack of $\mathrm{size}(3n+1) \geq \mathrm{size}(n) + 1$ — equivalently the carry pattern of base-2/base-3 interaction — is the Sturmian process of rotation number $\log_2 3$: the sequence $\mathrm{size}(3^k) - \lfloor k \log_2 3 \rfloor$ and its orbit-level analogues. The wall of Theorem 6.2 sits where the linear bit budget meets this slack; the cycle ceiling of §7 is the effective rational approximation of $\log_2 3$; the continuity no-go of Theorem 5.2 reflects that the carry process is invisible 3-adically. Effective results on this object are scarce; the only unconditional anchor we know is Stewart's theorem on digits in two bases [20] (see also [18]), and the Collatz process is known to embed base conversion [22].

Question 8.1 (carry Mason–Stothers, weak form)**.** Call an odd step $c \mapsto (3c+1)/2^{\tau}$ a *carry step* if $\mathrm{size}(3c+1) = \mathrm{size}(c) + 2$ (the excess case of the lossy size law). Is there an effective $\delta > 0$ and q_0 such that every admissible word with $q \geq q_0$ zero-steps has at least $\delta\, q$ carry steps among them? (The density must be measured against q: words consisting of `1`-letters only contain no odd step at all.) Equivalently: an effective positive lower density of carries in the base-2 Sturmian coding of multiplication by 3 along admissible orbits. Such a bound would sharpen the zero-edge size law and hence the bit budget of §6; whether it suffices to pass the bit-criterion wall ($Q \approx 7.7m$ on the faithful tower) from outside the linear class is part of the question.

8.2. **The $\mathbb{F}_2[x]$ twin, formalized end-to-end.** The control experiment makes the diagnosis precise. On $\mathbb{F}_2[x]$ the Collatz analogue of Hicks–Mullen–Yucas–Zavislak [9] and Snapp–Tracy [21] (see [3, 4, 2] for recent function-field developments, including general coefficient rings, completions, and sharp stopping times)

$$T(f) = f/x \quad (f(0) = 0), \qquad T(f) = (x+1)f + 1 \quad (f(0) = 1)$$

is a theorem: every nonzero f reaches 1. We formalized it completely (`F2Collatz.lean`; to our knowledge the first formalization), with the exact local laws displayed against their lossy integer twins:

$\mathbb{F}_2[x]$ (exact; Lean theorem)	$\mathbb{Z}$ (lossy; all difficulty here)
$\deg((x+1)f+1) = \deg f + 1$	$\operatorname{size}(3n+1) \geq \operatorname{size}(n) + 1$
$\deg(f/x) = \deg f - 1$	$\operatorname{size}(n/2^\tau) = \operatorname{size}(n) - \tau$
odd step lands even	$3n + 1$ even
$\deg(T^{[k]}f) \leq \deg f + 1\ \forall k$	*no integer counterpart*

(Lean names, row by row: `oddStep_natDegree`, `evenStep_natDegree`, `oddStep_coeff_zero`, `collatzT_iterate_natDegree_le`.)

Theorem 8.2 (the HMYZ theorem; `collatz_F2X`). *For every $f \in \mathbb{F}_2[x]$, $f \neq 0$, there is k with $T^{[k]}f = 1$.*

The proof is strong induction on the degree. The exit from a plateau (a run of odd iterates of constant degree) is governed by the identity (`plateau_step_identity`)

$$x\left(T^2 f + 1\right) \;=\; (x+1)\,(f+1):$$

in the $f + 1$ coordinate the two-step map is multiplication by $(x+1)/x$, so with $v = \operatorname{ord}_x(f_0 + 1) \leq \deg f_0$ the iterates $T^{2i} f_0$ $(i < v)$ are odd of degree $\deg f_0$, $T^{2v} f_0$ is even, and the degree drops at step $2v + 1$. This is the trailing-coefficient argument of Snapp and Tracy [21], who proved the theorem concurrently with [9]; `plateau_exit` formalizes its existential form (some iterate has strictly smaller degree) by induction on a bound for v, and the whole development is pure ring algebra.

The moral is sharpened by a falsity: the naive function-field analogue of Baker's theorem fails over $\mathbb{F}_2[x]$ (the Frobenius unit gaps $x^{2^k} + (x+1)^{2^k} = 1$ give infinitely many "small" gaps), and yet Theorem 8.2 stands. The integer difficulty is therefore *not* reducible to "effective Baker is missing": what the function field has and $\mathbb{Z}$ lacks is carry-freeness of the size function. Question 8.1 is the precise residue of the comparison.

9. Related work

Surveys and general background: [12, 13, 25]. Forward-density and stochastic results: [24, 7, 10, 23]; our `SyracuseMeasure.lean` formalizes the support analysis of the Syracuse 2-adic law used in [23]. Cycles: [6, 19, 8]; verification: [17, 1]. Undecidability of generalizations: [5, 11]. Function fields: [9, 3, 4, 2]; base conversion: [22]; two-base digit results: [18, 20]. Formalization: Lean 4 [14] and Mathlib [15]. Closest prior work on the cycle half: Merle's conditional Lean 4 exclusion of nontrivial cycles [16], which derives cycle bounds from convergents of $\log_2 3$ under Baker-type, computational-verification and product-bound hypotheses; our reduction takes only the verification frontier as a hypothesis, uses no Baker-type input, and is kernel-checked without compiled oracles. We are not aware of prior machine-checked obstruction results for potential-based Collatz strategies, nor of a prior formalization of the polynomial analogue [9, 21].

10. Methodology, evidence tiers, and axiom hygiene

The collaboration ran as a funnel with three evidence tiers, and we report every claim with its tier:

- **T1** *Lean theorems* over Mathlib, zero `sorry`, axiom set exactly {`propext`, `Classical.choice`, `Quot.sound`} (verified per headline theorem with `#print axioms`; Appendix A). All theorems cited in §§2–8 are T1.
- **T1′** Lean theorems whose proofs invoke the compiled evaluator (`native_decide`); these survive only in legacy exploratory strata and in the per-instance certificate family (`Cert_m1_*` and seven early exploratory modules, inventoried in `formalization.yaml`), where the trusted base includes the Lean compiler. A *machine-generated audit* (`AxiomAudit.lean`, regenerated by `tools/gen_formalization_yaml.py`) runs `#print axioms` on every declaration cited in this paper and verifies that each depends on at most the three standard axioms; thirteen former `native_decide` anchors were re-proved kernel-only during the programme.
- **T2** Exact-rational tool verdicts: LP Farkas certificates re-derived in exact arithmetic and re-verified against the model by an independent code path (the GATE-H sweep of §6; the $m = 1$ certificates and the feasibility witness promoted to T1).
- **T3** Numerical gates (profiles, frontiers, shape diagnostics) guiding the search; never load-bearing for a stated theorem.

The working method — a numerical gate first, adversarial verification second (two plausible intermediate claims produced by exploratory runs were *refuted* at this stage and never entered the corpus), Lean authentication last — kept the formal artifact small relative to the search space explored. The division of labour: the human author set the programme, the targets and the acceptance criteria; the AI assistant proposed and executed proofs, gates and refactors; everything that survived is machine-checked, so trust reduces to the Lean kernel plus the stated axioms, not to either author.

Reproducibility: the repository contains the full Lean source (65 modules), the gate scripts and their JSON outputs (`tools/`), and build instructions (narrow `lake build` targets; only the legacy T1′ certificates are slow). A tagged commit accompanies the arXiv version.

11. Open problems

(1) **CMS** (Question 8.1), weak and orbit forms — the named residue of the whole map. Any effective carry density extends the closed-form barrier past the wall.
(2) **The split crux**: a decomposition of level-$(m+1)$ paths whose split point is *not* required to be m-precise and whose crux survives the ladder witness of Theorem 3.4 (every m-precise split predicate is refuted).
(3) **Uniformity of the LP wall**: the GATE-H mechanism is identical at $m = 1, 2$; a uniform-in-m Lean theorem ("for every m, both classes die exactly at the bit criterion") looks within reach by hand.
(4) **The cycle ceiling**: any effective linear-form bound plugs into `EffectiveLinearForm`; short of that, pushing convergent pairs is mechanical but unbounded.
(5) **Beyond feature-linearity**: the obstruction map is design information — it says any viable potential must be non-linear in (size, j), 3-adically discontinuous, and immune to the deep-jump trap — in particular to the goal-adjacent trap edge $(j, 4) \to (j, 1)$. Characterizing the minimal such class is open.
(6) **Function-field bonus**: the Frobenius–Catalan classification $x^A + (x+1)^m = 1 \iff A = m = 2^k$ over $\mathbb{F}_2[x]$, a clean candidate for Mathlib.

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Appendix A. Axiom audit

Every declaration named in this paper is audited with `#print axioms` by the machine-generated module `AxiomAudit.lean` (regenerated from the manuscript itself by `tools/gen_formalization_yaml.py`; results recorded in `formalization.yaml` at the repository root). Each reports exactly {`propext`, `Classical.choice`, `Quot.sound`} — the three standard Lean axioms. The legacy modules that still use the compiled evaluator (tier T1′

of §10) are inventoried in the same file; nothing cited here depends on them. Selected audit rows:

theorem	module
`uniform_slope_barrier_of_block`	`KappaSplitReduction`
`not_firstMPrecisionSuffixPositive`	`LadderExists`
`not_blockBoundaryExists`	`LadderExists`
`invReachable_units`	`ReachUnits`
`no_decaying_modulus_corrector`	`AttractorNoGo`
`kappa_bounded_barrier_bitlen`, `kappa_barrier_m1_Q7`	`BitlenPotential`
`kappa_barrier_at_m2_Q16`, `kappa_barrier_at_m3_Q23`	`FaithfulBarrier`
`linear_size_potential_m1_Q7`	`HybridPotentialNoGo`
`no_linear_size_potential_m1_Q8`	`HybridPotentialNoGo`
`no_hybrid_potential_m1_Q8`	`HybridPotentialNoGo`
`syr_cycle_window`, `syr_master_corridor`	`CycleReduction`
`syr_no_cycle_below_16266`	`CycleReduction`
`collatz_no_cycle_below_359` (and 16266)	`CollatzBridge`
`collatz_F2X`, `plateau_exit`	`F2Collatz`

Appendix B. The Lean module map

The artifact comprises 65 modules. Main groups (full map in the repository `README`):

- graph model and word correspondence: `AdmissibleBasic`, `InverseGraph`, `S202Cylinders`, `PathWordCorrespondence`, `S212Forward`, `AnalyticBarrier`;
- the obstruction map: `KappaSplitReduction`, `LadderExists`, `LadderRefutation`, `ReachUnits`, `OneEdgeMod9`, `AttractorNoGo`, `HybridPotentialNoGo`;
- the positive frontier: `BitlenPotential`, `FaithfulBarrier`, `AS202Lift`;
- arithmetic infrastructure: `DiscreteLog`, `LTEBound`, `RhoTransitionLaws`, `BridgeTruth`, `ClosedWalk`;
- the cycle half: `CycleEquation`, `CycleReduction`, `CollatzBridge`;
- the function-field twin: `F2Collatz`; the measure-theoretic support analysis: `SyracuseMeasure`; the legacy certificates: `Cert_*` (tier T1′).

Build: `lake build CollatzLean4.<Module>` per module; the T1′ certificates are the only slow targets.

Appendix C. Numerical gates: scripts and data

The `tools/` directory (committed, with JSON outputs) contains the gate scripts behind the T2/T3 claims, including: `phi_realkam_lp.py` (the Hölder-corrector LP with ultrametric dendrogram hops and exact witness re-verification), `phi_hybrid_gate.py` (the GATE-H constraint-generation LP; exact rational Farkas certificates; the source of Theorems 6.2–6.3), `phi_hybrid_interior.py` (pole-set robustness), `phi_tropical_*.py` (the tropical refutation and its witness), `sturmian_gate.py`, and the ∼75 archived research probes whose verdicts shaped the programme. Each script documents the epistemic status of each possible outcome.

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